

1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of students who appeared in an exam in five different years 2020, 2021, 2022, 2023, and 2024.

The table given below shows the number of students who appeared in the exam, the ratio of male students to female students, who appeared in the exam, and the number of students, who cleared the exam out of the total number of male students who appeared in the exam, in each of the given years.

Year	No. of students who appeared in the exam	The ratio of male students to female students, who appeared in the exam	Percentage of male students, who cleared the exam out of the total number of male students who appeared in the exam
2020	6x	7: 5	30%
2021	Y	5: 8	15%
2022	5x	14: 11	24%
2023	y – 2000	5: 3	20%
2024	Z	17: 11	25%

Note: The calculated values of 'x', 'y', and 'z' will be the same for all the questions.

If the number of male students, who didn't clear the exam in 2022, is 6384, then find the difference between the number of female students, who appeared in the exam in 2020 and 2022?

- a. 1050
- b. 1200
- c. 750
- d. 900
- e. 1350

2. Questions

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Study the following data carefully and answer the questions:

The data given below is related to the number of students who appeared in an exam in five different years 2020, 2021, 2022, 2023, and 2024.

The table given below shows the number of students who appeared in the exam, the ratio of male students to female students, who appeared in the exam, and the number of students, who cleared the exam out of the total number of male students who appeared in the exam, in each of the given years.

Year	No. of students who appeared in the exam	The ratio of male students to female students, who appeared in the exam	Percentage of male students, who cleared the exam out of the total number of male students who appeared in the exam
2020	6x	7: 5	30%
2021	Y	5: 8	15%
2022	5x	14: 11	24%
2023	y – 2000	5: 3	20%
2024	Z	17: 11	25%

Note: The calculated values of ‘x’, ‘y’, and ‘z’ will be the same for all the questions.

If the ratio of the number of male students, who cleared the exam in 2020, to those, who cleared the exam in 2024, is 126: 85, then find the ratio of the total number of students, who appeared in the exam in 2022, to those, who appeared in the exam in 2024? (Refer from previous question)

- a. 5: 4
- b. 3: 2
- c. 24: 19
- d. 10: 7
- e. 15: 14

3. Questions

If 24% of the female students who appeared for the exam in 2020 cleared it, 25% of those who appeared in 2022 cleared it, and 30% of those who appeared in 2024 cleared it, then find the average number of female students who cleared the exam in 2020, 2022, and 2024.(Refer from previous questions)

- a. 1750
- b. 1675
- c. 1700
- d. 1720
- e. None of these

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4. Questions

If the number of male students, who cleared the exam in 2023, is 1500 more than those, who cleared the exam in 2021, then find that the total number of students, who appeared in the exam in 2021 is what percent more than those, who appeared in the exam in 2023?(Refer from previous questions)

- a. 12.5%

- b. 8.33%
- c. 5%
- d. 15%
- e. 3.5%

5. Questions

Find which of the following equations is/are true?

I: $y = 2z - x$

II: $y - z = 4x$

III: $y - x = z + 10000$

(Refer from previous questions)

- a. Only I and II
- b. Only II and III
- c. Only II
- d. Only I
- e. Only I and III

6. Questions

Study the following data carefully and answer the questions:

There are two mixtures A and B and each mixture contains different quantities [in Litre(L)] of cow's milk and buffalo's milk.

The cost of cow's milk is ₹10m per L, the cost of buffalo's milk ₹15m per L and the cost of mixture A is ₹11m per L. The difference between the quantities of cow's milk in mixtures A and B is 22 L and the quantity of buffalo's milk in mixture A is 7 L less than that in mixture B. The difference between the quantities of cow's milk and buffalo's milk in mixture A is 24 L. The cost of mixture B is more than ₹12m per L.

When 37.5% quantity of mixture A is mixed with the whole quantity of mixture B and then 20 L quantity of mixture B is taken out, then what will be the difference between the remaining quantities of cow's milk and buffalo's milk in mixture B?

- a. 2 L
- b. 4.5 L
- c. 1 L
- d. 5 L
- e. 2.5 L

7. Questions

If the cost of cow's milk and buffalo's milk increased by 400%, then find the difference between the cost of Mixture A and Mixture B.

- a. ₹5 per L
- b. ₹12 per L
- c. ₹8 per L
- d. ₹10 per L
- e. ₹15 per L

8. Questions

If 50% quantity of mixture A is mixed with the 60% quantity of mixture B and the quantities of cow's milk and buffalo's milk in the new mixture are 'M' L and 'N' L respectively, then find that which of the following statements is/are not true?

Note: the cost of cow's milk and the cost of buffalo's milk are taken from the previous question.

I: The ratio of M to N is 22: 15.

II: The cost of the new mixture is more than ₹55 per L.

III: The difference between M and N is a prime number.

- a. Only II and III
- b. Only I and III
- c. Only III
- d. Only I and II
- e. Only II

9. Questions

Inlet pipe A and outlet pipe C together can empty a filled tank in 30 minutes, while inlet pipe B and outlet pipe C together can empty the filled tank in 90 minutes. If pipe A is an inlet pipe and pipe B is an outlet pipe, how long will it take for pipes A and B together to empty the filled tank?

- a. Can't be determined
- b. 15 minutes
- c. 45 minutes
- d. 60 minutes
- e. 30 minutes

10. Questions

The difference between compound interest and simple interest on ₹(X + Y) at 12% rate after 2 years is ₹252 and the difference between compound interest and simple interest on ₹(X – Y) at 24% rate after 2 years is ₹720. Find which of the following statements is/are true?

I: The difference between compound interest and simple interest on ₹(X + 2Y) at 8% rate after 2 years is ₹168.

II: Y is 20% of X.

III: The compound interest received on ₹(X – Y) at 6% rate after 2 years is ₹1455.

- a. Only II
- b. Only I and III
- c. None is true
- d. Only II and III
- e. Only I

11. Questions

A and B started a business and invested in the ratio of 2: 1. After 't' months, A withdrew 33(1/3)% of his initial investment and after 3 more months, B invested 100% more of his initial investment.

At the end of the 10th month, the ratio of A's profit share to B's profit share is 25: 18. If the ratio of A's investment to B's investment were 5: 4 and the time periods of A and B were (t + 3) months and (t + 5) months respectively, then what would have been the ratio of A's profit share to B's profit share?

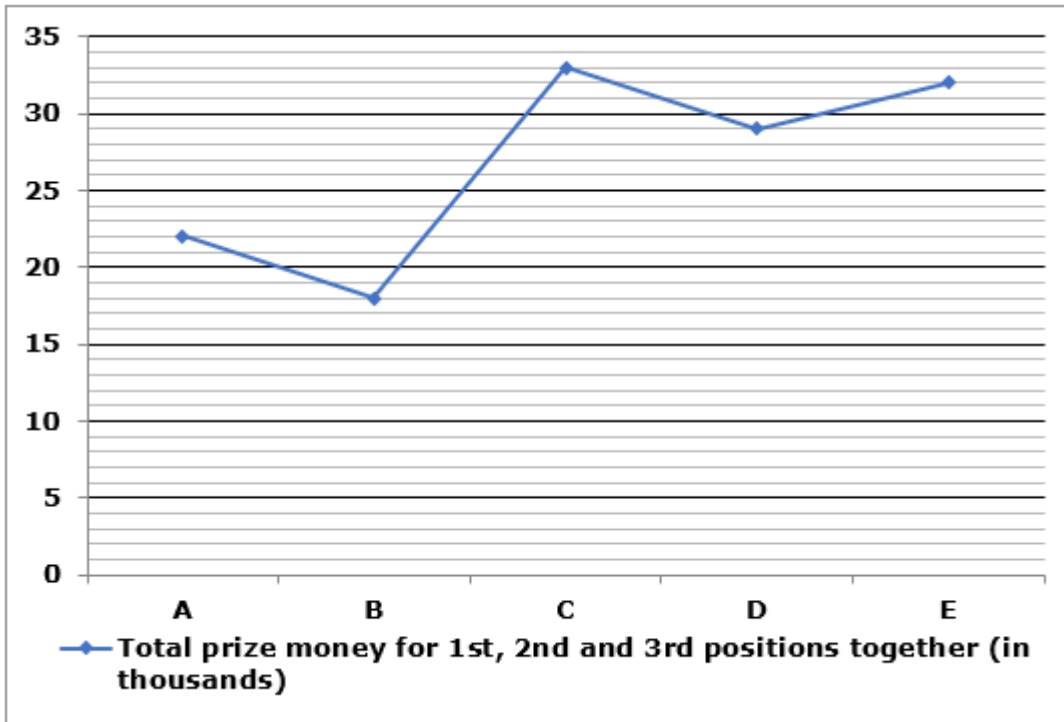
- a. 2: 3
- b. 5: 4
- c. 1: 2
- d. 4: 3
- e. 1: 1

12. Questions

Study the following data carefully and answer the questions: @QuizByMickey

Five games A, B, C, D, and E are organized in a school and the prize money for 1st, 2nd, and 3rd positions in each game is different. More than one candidate can be in a position.

The line graph given below shows the total prize money for 1st, 2nd, and 3rd positions together in each of the given games.



The table given below shows the following data:

Game	The prize money for 2 nd position as % of that for 1 st position	The prize money for 3 rd position as % of that for 1 st position
A	70%	50%
B	75%	50%
C	66(2/3)%	53(1/3)%
D	75%	66(2/3)%
E	62.5%	37.5%

If the number of candidates at 1st position in game C is 0.06% of the prize money for 1st position in game C, the number of candidates at 1st position in game D is 0.05% of the prize money for 1st position in game D and the number of candidates at 1st position in game E is 0.075% of the prize money for 1st position in game E, then find the total amount received by the number of candidates at 1st position of games C, D and E together.

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- ₹349000
- ₹409000
- ₹399000
- ₹419000
- ₹379000

13. Questions

If the average prize money for the 1st position of games A, B, and C is Rs.M and the average prize money for the 2nd position of games A, C, and E is Rs.N, then find the average prize money for 3rd position of games A, B and E.

- a. ₹(M – N + 1500)
- b. ₹(M – N + 2500)
- c. ₹(M – N + 3000)
- d. ₹(M – N + 500)
- e. None of these

14. Questions

Which of the following statements is/are not true?

I: The ratio of prize money for 2nd position in game B to that in game D is 2: 3.

II: The average prize money for 3rd positions in each of the given games is ₹6100.

III: The prize money for 1st position in game C is 87.5% more than that in game B.

- a. Only I and II
- b. Only III
- c. Only II and III
- d. Only II
- e. Only I

15. Questions

If the number of candidates at 1st position in game A is 3, the number of candidates at 2nd position in game A is 6 and the number of candidates at 3rd position in game A is 9, then find the total of prize money distributed for 1st, 2nd and 3rd positions of game A?

- a. ₹133000
- b. ₹125000
- c. ₹136000
- d. ₹152000
- e. ₹117000

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16. Questions

If the prize money for 1st position in game B were 12.5% more than the original prize money, the

prize money for 2nd position in game B was $33\frac{1}{3}\%$ more than the original prize money and the prize money for 3rd position in game B remains same, then the new sum of prize money for 1st, 2nd and 3rd positions in game B was what percent more/less than the original sum?

- a. 12.5%
- b. 20%
- c. $16\frac{2}{3}\%$
- d. $11\frac{1}{9}\%$
- e. 7.5%

17. Questions

In each question, three equations I, II, and III are given. You have to solve all the three equations and give an answer.

If we take the smaller root of the following equations, find the relationship among those roots.

I: $x^2 + 7x - 120 = 0$

II: $y^2 - 27y + 180 = 0$

III: $z^2 - 39z + 350 = 0$

- a. $x < z < y$
- b. $z < y < x$
- c. $z < x < y$
- d. $x < y < z$
- e. $y < z < x$

18. Questions

If we take any one of the roots from the three equations and compare the options.

I: $x^2 - 7x - 60 = 0$

II: $y^2 + 9y - 90 = 0$

III: $z^2 - 14z + 40 = 0$

Find that which of the following can't be true?

- a. $x > y > z$
- b. $y > z > x$
- c. $y > x > z$

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- d. $x > z > y$
- e. All can be correct

19. Questions

I: $x^2 - x - 2A = 0$

II: $y^2 - y - 6B = 0$

III: $z^2 - (A - B)z - 24 = 0$

One root of 'x' is 5 and one root of 'y' is -5.

Find that which of the followings are true.

P: The smallest root of 'x' > the smallest root of 'y'

Q: The largest root of 'y' > the smallest root of 'z'

R: The largest root of 'z' > the largest root of 'x'

- a. Only P and Q
- b. Only R
- c. Only P and R
- d. All are true
- e. Only P

20. Questions

I: $3x + y = 19$

II: $2y + 7z = 35$

III: $5x + 3z = 29$

- a. $x > y > z$
- b. $y > x > z$
- c. $y > z > x$
- d. $x > z > y$
- e. $z > x > y$

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Explanations:

1. Questions

Answer: D

Since, the number of male students, who didn't clear the exam in 2022 = 6384

So, the number of male students, who appeared in the exam in 2022 = $6384 * [100/(100 - 24)] = 8400$

And the number of female students, who appeared in the exam in 2022 = $8400 * (11/14) = 6600$

Since, the total number of students, who appeared in the exam in 2022 = $8400 + 6600 = 15000$

So,

$$5x = 8400 + 6600$$

$$x = 3000$$

Now, the total number of students, who appeared in the exam in 2020 = $6x = 6 * 3000 = 18000$

And the number of female students, who appeared in the exam in 2020 = $18000 * (5/12) = 7500$

So, the required difference = $7500 - 6600 = 900$

2. Questions

Answer: E

Since, the total number of students, who appeared in the exam in 2020 = $6x = 6 * 3000 = 18000$

So, the number of male students, who appeared in the exam in 2020 = $18000 * (7/12) = 10500$

And the number of male students, who cleared the exam in 2020 = $30\% \text{ of } 10500 = 3150$

Since, the ratio of the number of male students, who cleared the exam in 2020, to those, who cleared the exam in 2024 = 126: 85

So, the number of male students, who cleared the exam in 2024 = $3150 * (85/126) = 2125$

The number of male students, who appeared in the exam in 2024 = $2125 * (100/25) = 8500$

The number of female students, who appeared in the exam in 2024 = $8500 * (11/17) = 5500$

And the total number of students, who appeared in the exam in 2024 = $8500 + 5500 = 14000$

So,

$$z = 14000$$

Since, the total number of students, who appeared in the exam in 2022 = $5x = 5 * 3000 = 15000$

So, the required ratio = $15000: 14000 = 15: 14$

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3. Questions

Answer: C

Since, the total number of students, who appeared in the exam in 2020 = $6x = 6 * 3000 = 18000$

So, the number of female students, who appeared in the exam in 2020 = $18000 * (5/12) = 7500$

And the number of female students, who cleared the exam in 2020 = $24\% \text{ of } 7500 = 1800$

Since, the total number of students, who appeared in the exam in 2022 = $5x = 5 * 3000 = 15000$

So, the number of female students, who appeared in the exam in 2022 = $15000 * (11/25) = 6600$

And the number of female students, who cleared the exam in 2022 = 25% of 6600 = 1650

Since, the total number of students, who appeared in the exam in 2024 = $z = 14000$

So, the number of female students, who appeared in the exam in 2024 = $14000 * (11/28) = 5500$

And the number of female students, who cleared the exam in 2024 = 30% of 5500 = 1650

Required average = $(1800 + 1650 + 1650)/3 = 1700$

4. Questions

Answer: B

Since, the total number of students, who appeared in the exam in 2021 = y

So, the number of male students, who appeared in the exam in 2021 = $y * (5/13) = (5y/13)$

And the number of male students, who cleared the exam in 2021 = 15% of $(5y/13) = (5y/13) * (15/100) = (3y/52)$

Since, the total number of students, who appeared in the exam in 2023 = $(y - 2000)$

So, the number of male students, who appeared in the exam in 2023 = $(y - 2000) * (5/8) = (5y - 10000)/8$

And the number of male students, who cleared the exam in 2023 = 20% of $(5y - 10000)/8 = [(5y - 10000)/8] * (20/100) = (y - 2000)/8$

From the question:

$$[(y - 2000)/8] - (3y/52) = 1500$$

$$(13y - 26000 - 6y)/104 = 1500$$

$$7y = 182000$$

$$y = 26000$$

Since, the total number of students, who appeared in the exam in 2021 = 26000

And the total number of students, who appeared in the exam in 2023 = $26000 - 2000 = 24000$

So, the required percentage = $[(26000 - 24000)/24000] * 100 = 8.33\%$

5. Questions

Answer: C

$$x = 3000, y = 26000, z = 14000$$

From I:

$$y = 2z - x$$

$$26000 = 2 * 14000 - 3000$$

$$26000 \neq 25000$$

So, I is not true.

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From II:

$$y - z = 4x$$

$$26000 - 14000 = 4 * 3000$$

$$12000 = 12000$$

So, II is true.

From III:

$$y - x = z + 10000$$

$$26000 - 3000 = 14000 + 10000$$

$$23000 \neq 24000$$

So, III is not true.

Hence, only II is true.

6. Questions

The cost of cow's milk = ₹10m per L

The cost of buffalo's milk = ₹15m per L

The cost of mixture A = ₹11m per L

Let the quantities of cow's milk and buffalo's milk in mixture A are 'x' L and 'y' L respectively.

So,

$$[(x * 10m) + (y * 15m)] / (x + y) = 11m$$

$$10x + 15y = 11x + 11y$$

$$x = 4y$$

$$x : y = 4 : 1$$

Since, the difference between the quantities of cow's milk and buffalo's milk in mixture A = 24 L

So, the quantity of cow's milk in mixture A = $24 * [4/(4 - 1)] = 32$ L

And the quantity of buffalo's milk in mixture A = $24 * [1/(4 - 1)] = 8$ L

Since the quantity of buffalo's milk in mixture A is 7 L less than that in mixture B.

So, the quantity of buffalo's milk in mixture B = $8 + 7 = 15$ L

Since, the difference between the quantities of cow's milk in mixtures A and B = 22 L

So, the quantity of cow's milk in mixture B will either be $32 - 22 = 10$ L or $32 + 22 = 54$ L

Case I: When the quantity of cow's milk in mixture B = 10 L

So, the cost of mixture B = $[(10 * 10m) + (15 * 15m)] / (10 + 15) = ₹13m$ per L

Case II: When the quantity of cow's milk in mixture B = 54 L

So, the cost of mixture B = $[(54 * 10m) + (15 * 15m)] / (54 + 15) = ₹11m$ per L (approx.)

Since the cost of mixture B is more than ₹12m per L.

So, case II can be neglected.

So, the quantity of cow's milk in mixture B = 10 L

And the quantity of buffalo's milk in mixture B = 15 L

Answer: A

Let according to the question

The total quantity of mixture A = 40 L

The ratio of cow's milk to buffalo's milk in mixture A = 32: 8 = 4: 1

When 37.5% quantity of mixture A is mixed with the whole quantity of mixture B:

The quantity of cow's milk in 37.5% quantity of mixture A = 37.5% of 40 * (4/5) = 12 L

The quantity of buffalo's milk in 37.5% quantity of mixture A = 37.5% of 40 * (1/5) = 3 L

The new quantity of cow's milk in mixture B = 10 + 12 = 22 L

The new quantity of buffalo's milk in mixture B = 15 + 3 = 18 L

The new ratio of cow's milk to buffalo's milk in mixture B = 22: 18 = 11: 9

When 20 L quantity of mixture B is taken out:

The remaining quantity of cow's milk in mixture B = 22 - 20 * (11/20) = 11 L

The remaining quantity of buffalo's milk in mixture B = 18 - 20 * (9/20) = 9 L

So, the required difference = 11 - 9 = 2 L

7. Questions

The cost of cow's milk = ₹10m per L

The cost of buffalo's milk = ₹15m per L

The cost of mixture A = ₹11m per L

Let the quantities of cow's milk and buffalo's milk in mixture A are 'x' L and 'y' L respectively.

So,

$$[(x * 10m) + (y * 15m)] / (x + y) = 11m$$

$$10x + 15y = 11x + 11y$$

$$x = 4y$$

$$x: y = 4: 1$$

Since, the difference between the quantities of cow's milk and buffalo's milk in mixture A = 24 L

So, the quantity of cow's milk in mixture A = 24 * $[4 / (4 - 1)]$ = 32 L

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And the quantity of buffalo's milk in mixture A = $24 * [1/(4 - 1)] = 8$ L

Since the quantity of buffalo's milk in mixture A is 7 L less than that in mixture B.

So, the quantity of buffalo's milk in mixture B = $8 + 7 = 15$ L

Since, the difference between the quantities of cow's milk in mixtures A and B = 22 L

So, the quantity of cow's milk in mixture B will either be $32 - 22 = 10$ L or $32 + 22 = 54$ L

Case I: When the quantity of cow's milk in mixture B = 10 L

So, the cost of mixture B = $[(10 * 10m) + (15 * 15m)]/(10 + 15) = ₹13m$ per L

Case II: When the quantity of cow's milk in mixture B = 54 L

So, the cost of mixture B = $[(54 * 10m) + (15 * 15m)]/(54 + 15) = ₹11m$ per L (approx.)

Since the cost of mixture B is more than ₹12m per L.

So, case II can be neglected.

So, the quantity of cow's milk in mixture B = 10 L

And the quantity of buffalo's milk in mixture B = 15 L

Answer: D

$m = 1$

The cost of cow's milk = ₹10m = ₹10

The cost of buffalo's milk = ₹15m = ₹15

Now,

The cost of cow's milk = $10 * 500/100 = ₹50$ per L

The cost of buffalo's milk = $15 * 500/100 = ₹75$ per L

Since, the quantity of cow's milk in mixture A = 32 L

And the quantity of buffalo's milk in mixture A = 8 L

So, the cost of mixture A = $[(32 * 50) + (8 * 75)]/(32 + 8) = ₹55$ per L

Since, the quantity of cow's milk in mixture B = 10 L

And the quantity of buffalo's milk in mixture B = 15 L

So, the cost of mixture B = $[(10 * 50) + (15 * 75)]/(10 + 15) = ₹65$ per L

Required difference = $65 - 55 = ₹10$ per L

8. Questions

The cost of cow's milk = ₹10m per L

The cost of buffalo's milk = ₹15m per L

The cost of mixture A = ₹11m per L

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Let the quantities of cow's milk and buffalo's milk in mixture A are 'x' L and 'y' L respectively.

So,

$$[(x * 10m) + (y * 15m)]/(x + y) = 11m$$

$$10x + 15y = 11x + 11y$$

$$x = 4y$$

$$x : y = 4 : 1$$

Since, the difference between the quantities of cow's milk and buffalo's milk in mixture A = 24 L

$$\text{So, the quantity of cow's milk in mixture A} = 24 * [4/(4 - 1)] = 32 \text{ L}$$

$$\text{And the quantity of buffalo's milk in mixture A} = 24 * [1/(4 - 1)] = 8 \text{ L}$$

Since the quantity of buffalo's milk in mixture A is 7 L less than that in mixture B.

$$\text{So, the quantity of buffalo's milk in mixture B} = 8 + 7 = 15 \text{ L}$$

Since, the difference between the quantities of cow's milk in mixtures A and B = 22 L

$$\text{So, the quantity of cow's milk in mixture B will either be } 32 - 22 = 10 \text{ L or } 32 + 22 = 54 \text{ L}$$

Case I: When the quantity of cow's milk in mixture B = 10 L

$$\text{So, the cost of mixture B} = [(10 * 10m) + (15 * 15m)]/(10 + 15) = ₹13m \text{ per L}$$

Case II: When the quantity of cow's milk in mixture B = 54 L

$$\text{So, the cost of mixture B} = [(54 * 10m) + (15 * 15m)]/(54 + 15) = ₹11m \text{ per L (approx.)}$$

Since the cost of mixture B is more than ₹12m per L.

So, case II can be neglected.

$$\text{So, the quantity of cow's milk in mixture B} = 10 \text{ L}$$

$$\text{And the quantity of buffalo's milk in mixture B} = 15 \text{ L}$$

Answer: B

Since, the ratio of cow's milk to buffalo's milk in mixture A = 32: 8 = 4: 1

$$\text{So, the quantity of cow's milk in 50\% quantity of mixture A} = 50\% \text{ of } 40 * (4/5) = 16 \text{ L}$$

$$\text{And the quantity of buffalo's milk in 50\% quantity of mixture A} = 50\% \text{ of } 40 * (1/5) = 4 \text{ L}$$

Since, the ratio of cow's milk to buffalo's milk in mixture B = 10: 15 = 2: 3

$$\text{So, the quantity of cow's milk in 60\% quantity of mixture B} = 60\% \text{ of } 25 * (2/5) = 6 \text{ L}$$

$$\text{And the quantity of buffalo's milk in 60\% quantity of mixture B} = 60\% \text{ of } 25 * (3/5) = 9 \text{ L}$$

$$\text{Now, the quantity of cow's milk in the new mixture} = M = 16 + 6 = 22 \text{ L}$$

$$\text{And the quantity of buffalo's milk in the new mixture} = N = 4 + 9 = 13 \text{ L}$$

From I:

Since, the ratio of M to N = M: N = 22: 13

So, I is not true.

From II:

Since, the cost of the new mixture:

$$[(22 * 50) + (13 * 75)] / (22 + 13)$$

$$[1100 + 975] / 35$$

₹59.3 per L (approx.)

So, II is true.

From III:

The difference of M and N = M – N = 22 – 13 = 9

Since 9 is not a prime number.

So, III is not true.

Hence, I and III are not true.

9. Questions

Answer: C

Since, inlet pipe A and outlet pipe C can empty the filled tank in 30 minutes.

So,

$$(1/A) - (1/C) = (1/30)$$

$$(1/C) = (1/A) - (1/30) \text{ -----(1)}$$

Since inlet pipe B and outlet pipe C can empty the filled tank in 90 minutes.

So,

$$(1/B) - (1/C) = (1/90)$$

$$(1/C) = (1/B) - (1/90) \text{ -----(2)}$$

From equations (1) and (2):

$$(1/A) - (1/30) = (1/B) - (1/90)$$

$$(1/A) - (1/B) = (1/30) - (1/90)$$

$$(1/A) - (1/B) = (3 - 1)/90$$

$$(1/A) - (1/B) = 1/45$$

If pipe A were an inlet pipe and pipe B were an outlet pipe, then at the time, in which pipes A and B together emptied the filled tank, would have been 45 minutes.

10. Questions

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Answer: C

Since, the difference between CI and SI on ₹(X + Y) at 12% rate after 2 years = ₹252

So,

$$(X + Y) * (12/100)^2 = 252$$

$$X + Y = 17500 \text{ -----(1)}$$

Since, the difference between CI and SI on ₹(X – Y) at 24% rate after 2 years = ₹720

So,

$$(X - Y) * (24/100)^2 = 720$$

$$X - Y = 12500 \text{ -----(2)}$$

From equations (1) + (2):

$$X + Y + X - Y = 17500 + 12500$$

$$X = 15000$$

From equation (1):

$$15000 + Y = 17500$$

$$Y = 2500$$

From I:

The difference between CI and SI on ₹(X + 2Y) at 8% rate after 2 years:

$$(X + 2Y) * (8/100)^2$$

$$= (15000 + 5000) * (8/100)^2$$

$$= ₹128$$

So, I is not true.

From II:

Since $Y = (2500/15000) * 100 = 16(2/3)\%$ of X.

So, II is not true.

From III:

The CI received on ₹(X – Y) at 6% rate after 2 years:

$$(X - Y) * (100 + 6/100)^2 - (X - Y)$$

$$= (15000 - 2500) * (106/100)^2 - (15000 - 2500)$$

$$= ₹1545$$

So, III is not true.

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Hence, none is true.

11. Questions

Answer: E

Let the initial investments of A and B be Rs. '6x' and Rs. '3x' respectively.

So, A's investment after 't' months = $66\frac{2}{3}\%$ of $6x = \text{Rs. } 4x$

And B's investment after (t + 3) months = 200% of $3x = \text{Rs. } 6x$

So, the ratio of A's profit share to B's profit share at the end of the 10th month:

$$= [6x * t + 4x * (10 - t)] : [3x * (t + 3) + 6x * (10 - t - 3)]$$

$$= [6t + 40 - 4t] : [3t + 9 + 42 - 6t]$$

$$= [2t + 40] : [51 - 3t]$$

Since, the ratio of A's profit share to B's profit share = 25: 18

So,

$$(2t + 40)/(51 - 3t) = 25/18$$

$$(2t + 40)/(17 - t) = 25/6$$

$$12t + 240 = 425 - 25t$$

$$37t = 185$$

$$t = 5$$

Now, if the ratio of A's investment to B's investment were 5: 4 and the time periods of A and B were (t + 3) months and (t + 5) months respectively.

So, the ratio of A's profit share to B's profit share would have been = $[5 * (t + 3)] : [4 * (t + 5)]$

$$= [5 * (5 + 3)] : [4 * (5 + 5)]$$

$$= 40: 40$$

$$= 1: 1$$

12. Questions

Game A:

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Let the prize money for 1st position = ₹10x

So, the prize money for 2nd position = 70% of $10x = ₹7x$

And the prize money for 3rd position = 50% of $10x = ₹5x$

Since the total prize money for all the three positions together = 22000

So,

$$10x + 7x + 5x = 22000$$

$$x = 1000$$

The prize money for 1st position = $10 * 1000 = ₹10000$

The prize money for 2nd position = $7 * 1000 = ₹7000$

The prize money for 3rd position = $5 * 1000 = ₹5000$

Similarly, we can calculate the data for other games as well:

Game	The prize money for 1 st position (in ₹)	The prize money for 2 nd position (in ₹)	The prize money for 3 rd position (in ₹)
A	10000	7000	5000
B	8000	6000	4000
C	15000	10000	8000
D	12000	9000	8000
E	16000	10000	6000

Answer: C

Since, the prize money for 1st position in game C = 15000

So, the number of candidates at 1st position in game C = 0.06% of 15000 = 9

Since, the prize money for 1st position in game D = 12000

So, the number of candidates at 1st position in game D = 0.05% of 12000 = 6

Since, the prize money for 1st position in game E = 16000

So, the number of candidates at 1st position in game E = 0.075% of 16000 = 12

Now, the total amount received by the number of candidates at 1st position of games C, D, and E together = $(15000 * 9) + (12000 * 6) + (16000 * 12)$

$$= 135000 + 72000 + 192000$$

$$= ₹399000$$

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13. Questions

Game A:

Let the prize money for 1st position = ₹10x

So, the prize money for 2nd position = 70% of 10x = ₹7x

And the prize money for 3rd position = 50% of 10x = ₹5x

Since the total prize money for all the three positions together = 22000

So,

$$10x + 7x + 5x = 22000$$

$$x = 1000$$

The prize money for 1st position = 10 * 1000 = ₹10000

The prize money for 2nd position = 7 * 1000 = ₹7000

The prize money for 3rd position = 5 * 1000 = ₹5000

Similarly, we can calculate the data for other games as well:

Game	The prize money for 1 st position (in ₹)	The prize money for 2 nd position (in ₹)	The prize money for 3 rd position (in ₹)
A	10000	7000	5000
B	8000	6000	4000
C	15000	10000	8000
D	12000	9000	8000
E	16000	10000	6000

Answer: C

Since, the average prize money for 1st position of games A, B and C = (10000 + 8000 + 15000)/3 = ₹11000

$$\text{So, } M = 11000$$

Since, the average prize money for 2nd position of games A, C and E = (7000 + 10000 + 10000)/3 = ₹9000

$$\text{So, } N = 9000$$

Now, the average prize money for 3rd position of games A, B and E = (5000 + 4000 + 6000)/3 = ₹5000 = ₹(M - N + 3000)

14. Questions

Game A:

Let the prize money for 1st position = ₹10x

So, the prize money for 2nd position = 70% of 10x = ₹7x

And the prize money for 3rd position = 50% of 10x = ₹5x

Since the total prize money for all the three positions together = 22000

So,

$$10x + 7x + 5x = 22000$$

$$x = 1000$$

The prize money for 1st position = $10 * 1000 = ₹10000$

The prize money for 2nd position = $7 * 1000 = ₹7000$

The prize money for 3rd position = $5 * 1000 = ₹5000$

Similarly, we can calculate the data for other games as well:

Game	The prize money for 1 st position (in ₹)	The prize money for 2 nd position (in ₹)	The prize money for 3 rd position (in ₹)
A	10000	7000	5000
B	8000	6000	4000
C	15000	10000	8000
D	12000	9000	8000
E	16000	10000	6000

Answer: D

From I:

Ratio of prize money for 2nd position in game B to that in game D = $6000: 9000 = 2: 3$

So, I is true.

From II:

The average prize money for 3rd position in each of the given games:

$$(5000 + 4000 + 8000 + 8000 + 6000)/5$$

$$₹6200$$

So, II is not true.

From III:

The prize money for 1st position in game B = ₹8000

The prize money for 1st position in game C = ₹15000

$$\text{Required percentage} = [(15000 - 8000)/8000] * 100 = 87.5\%$$

So, III is true.

Hence, only II is not true.

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15. Questions

Game A:

Let the prize money for 1st position = ₹10x

So, the prize money for 2nd position = 70% of 10x = ₹7x

And the prize money for 3rd position = 50% of 10x = ₹5x

Since the total prize money for all the three positions together = 22000

So,

$$10x + 7x + 5x = 22000$$

$$x = 1000$$

The prize money for 1st position = 10 * 1000 = ₹10000

The prize money for 2nd position = 7 * 1000 = ₹7000

The prize money for 3rd position = 5 * 1000 = ₹5000

Similarly, we can calculate the data for other games as well:

Game	The prize money for 1 st position (in ₹)	The prize money for 2 nd position (in ₹)	The prize money for 3 rd position (in ₹)
A	10000	7000	5000
B	8000	6000	4000
C	15000	10000	8000
D	12000	9000	8000
E	16000	10000	6000

Answer: E

Since, the prize money for 1st position of game A = ₹10000

So, the total prize money distributed for 1st position of game A = 10000 * 3 = ₹30000

Since, the prize money for 2nd position of game A = ₹7000

So, the total prize money distributed for 2nd position of game A = 7000 * 6 = ₹42000

Since, the prize money for 3rd position of game A = ₹5000

So, the total prize money distributed for 3rd position of game A = 5000 * 9 = ₹45000

Required total = (30000 + 42000 + 45000) = ₹117000

16. Questions

Game A:

Let the prize money for 1st position = ₹10x

So, the prize money for 2nd position = 70% of 10x = ₹7x

And the prize money for 3rd position = 50% of 10x = ₹5x

Since the total prize money for all the three positions together = 22000

So,

$$10x + 7x + 5x = 22000$$

$$x = 1000$$

The prize money for 1st position = 10 * 1000 = ₹10000

The prize money for 2nd position = 7 * 1000 = ₹7000

The prize money for 3rd position = 5 * 1000 = ₹5000

Similarly, we can calculate the data for other games as well:

Game	The prize money for 1 st position (in ₹)	The prize money for 2 nd position (in ₹)	The prize money for 3 rd position (in ₹)
A	10000	7000	5000
B	8000	6000	4000
C	15000	10000	8000
D	12000	9000	8000
E	16000	10000	6000

Answer: C

Since, the original prize money for 1st position in game B = ₹8000

So, the new prize money for 1st position in game B = 112.5% of 8000 = ₹9000

Since, the original prize money for 2nd position in game B = ₹6000

So, the new prize money for 2nd position in game B = 133(2/3)% of 6000 = ₹8000

Since, the prize money for 3rd position in game B = ₹4000

Since, the original sum of prize money for 1st, 2nd and 3rd positions in game B = 8000 + 6000 + 4000 = ₹18000

And the new sum of prize money for 1st, 2nd and 3rd position in game B = 9000 + 8000 + 4000 = ₹21000

So, the required percentage = $[(21000 - 18000)/18000] * 100 = 16(2/3)\%$

17. Questions

Answer: D

From equation I:

$$x^2 + 7x - 120 = 0$$

$$x^2 + 15x - 8x - 120 = 0$$

$$x(x + 15) - 8(x + 15) = 0$$

$$x = -15, 8$$

The smaller root = -15

From equation II:

$$y^2 - 27y + 180 = 0$$

$$y^2 - 15y - 12y + 180 = 0$$

$$y(y - 15) - 12(y - 15) = 0$$

$$y = 15, 12$$

The smaller root = 12

From equation III:

$$z^2 - 39z + 350 = 0$$

$$z^2 - 25z - 14z + 350 = 0$$

$$z(z - 25) - 14(z - 25) = 0$$

$$z = 25, 14$$

The smaller root = 14

$$(x = -15) < (y = 12) < (z = 14)$$

So, $x < y < z$

18. Questions

Answer: C

From equation I:

$$x^2 - 7x - 60 = 0$$

$$x^2 - 12x + 5x - 60 = 0$$

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$$x(x - 12) + 5(x - 12) = 0$$

$$x = 12, -5$$

From equation II:

$$y^2 + 9y - 90 = 0$$

$$y^2 + 15y - 6y - 90 = 0$$

$$(y + 15) - 6(y + 15) = 0$$

$$y = -15, 6$$

From equation III:

$$z^2 - 14z + 40 = 0$$

$$z^2 - 10z - 4z + 40 = 0$$

$$z(z - 10) - 4(z - 10) = 0$$

$$z = 10, 4$$

$x > y > z$ will be true only when, $x = 12$, $y = 6$ and $z = 4$

$y > z > x$ will be true only when, $y = 6$, $z = 4$ and $x = -5$

$x > z > y$ will be true, when, $x = 12$, $z = 10$ and $y = 6$ or -15 and when, $x = 12$, $z = 4$ and $y = -15$

$y > x > z$ will be true, when, $y = 6$, $x = -5$ and $z = 10$ and 4 are more than x

Since, the smaller root of 'y' is smaller than all roots of 'x' and 'z', so if we take, $y = 6$, then we will have to take, $x = -5$. Since -5 is smaller than both the roots of 'z', so $y > x > z$ can't be true.

19. Questions

Answer: D

From equation I:

$$x^2 - x - 2A = 0$$

Since one root of 'x' is 5.

$$\text{So, } 5^2 - 5 - 2A = 0$$

$$2A = 20$$

$$A = 10$$

$$x^2 - x - 2 * 10 = 0$$

$$x^2 - 5x + 4x - 20 = 0$$

$$x(x - 5) + 4(x - 5) = 0$$

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$$x = 5, -4$$

From equation II:

$$y^2 - y - 6B = 0$$

Since, one root of 'y' is -5.

$$\text{So, } (-5)^2 - (-5) - 6B = 0$$

$$6B = 25 + 5$$

$$B = 5$$

$$y^2 - y - 6 * 5 = 0$$

$$y^2 - 6y + 5y - 30 = 0$$

$$y(y - 6) + 5(y - 6) = 0$$

$$y = 6, -5$$

From equation III:

$$z^2 - (A - B)z - 24 = 0$$

$$z^2 - (10 - 5)z - 24 = 0$$

$$z^2 - 5z - 24 = 0$$

$$z^2 - 8z + 3z - 24 = 0$$

$$z(z - 8) + 3(z - 8) = 0$$

$$z = 8, -3$$

From P:

When, the smallest root of 'x' = -4 and the smallest root of 'y' = -5, then, $x > y$ will be true.

From Q:

When, the largest root of 'y' = 6 and the smallest root of 'z' = -3, then, $y > z$ will be true.

From R:

When, the largest root of 'z' = 8 and the largest root of 'x' = 5, then, $z > x$ will be true.

When, $z = 8$ and $x = 5$, then $z > x$ will be true.

Hence, all can be true.

20. Questions

Answer: B

$$3x + y = 19 \text{ -----(I)}$$

$$2y + 7z = 35 \text{ -----(II)}$$

$$5x + 3z = 29 \text{ -----(III)}$$

By equation (I) + (II) + (III):

$$(3x + y) + (2y + 7z) + (5x + 3z) = 19 + 35 + 29$$

$$8x + 3y + 10z = 83 \text{ -----(IV)}$$

By equation (IV) – 3 * (I):

$$8x + 3y + 10z - 9x - 3y = 83 - 3 * 19$$

$$10z - x = 26$$

$$x = 10z - 26 \text{ -----(V)}$$

From equations (III) and (V):

$$5(10z - 26) + 3z = 29$$

$$50z - 130 + 3z = 29$$

$$z = 3$$

From equation (II):

$$2y + 7 * 3 = 35$$

$$y = 7$$

From equation (I):

$$3x + 7 = 19$$

$$x = 4$$

Hence, $y > x > z$

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